

Cohomological Support Varieties for Monomial Ideals.

- Fauer, Fagerstrom, Katz, Stern, and Watson.

R - Noetherian local ring (or positively graded ring).

$\hat{R} \cong \mathbb{Q}/I$, $I \subseteq \mathfrak{m}^2$. $(\mathbb{Q}, \mathfrak{m}, k)$ - KLR, $k = \bar{k}$.

$I/\mathfrak{m}I$, $\eta = \mu(I)$

SII

k^n

$\longleftrightarrow \mathbb{A}_k^n$

$(f : f_1, \dots, f_n)$

• $[f] := f + \mathfrak{m}I$, class of f in $I/\mathfrak{m}I$.

• $\hat{M}$ as a $\hat{R}$ -module, given any $f \in I$, then $\hat{M}$ is a $\mathbb{Q}/(f)$ -module.

- Let M be a f.g. R -module.

The CSV of M is given by.

$$\mathbb{V}_R(M) := \{ [f] \in I/\mathfrak{m}I \mid [f] = 0 \text{ (or) } \text{pdim}_{\mathbb{Q}/(f)}(\hat{M}) = \infty \}$$

(equivalently) $f_a := \tilde{a}_1 f_1 + \dots + \tilde{a}_n f_n$, $\tilde{a}_i \in \mathbb{Q}$.

$$\Rightarrow \mathbb{V}_R(M) := \{ \underline{a} \in \mathbb{A}_k^n \mid \underline{a} = 0 \text{ (or) } \text{pdim}_{\mathbb{Q}/(f_a)}(\hat{M}) = \infty \}$$

$D(R)$ - derived Category of R .
 $m \in D^f(R)$ - Complexes with f.g homology

Alternate:

$$V_R(m) := \text{Supp}_S(\text{Ext}_E(m, K))$$

E - Koszul complex on $(f_1, \dots, f_n)$.

$K[x_1, \dots, x_n] =: S$ - Ring of Cohomological Operators.

• $\text{Ext}_E(m, K)$ is a f.g S -module.

By Nullstellensatz,

$$\sqrt{\text{ann}_S(\text{Ext}_E(m, K))} = \bigcap_{\underline{a} \in V_R(m)} (x_1 - a_1, \dots, x_n - a_n)$$

• $V_R(m)$ - are conical affine Varieties.

- [Pollitz, 19]

R is a Complete Intersection

$$\Leftrightarrow V_R(R) = \{0\}.$$

$\Rightarrow f \in I$, $(f, f_2, \dots, f_n) =: I$ regular sequence.

$\text{Kos}_{Q_{V(f)}}(f_2, \dots, f_n)$ finite. $\text{pdim}_{Q_{V(f)}}(\hat{R}) < \infty$.

$$\Rightarrow V_R(R) = \{0\}$$

- $R \rightsquigarrow Q/I \cong \hat{R}$ K of (Q, m, K) .

$$V_R(K) = A_K^n, \quad n = \mu(I).$$

$f_a \in I \subseteq m^2$, $Q/(f)$ is not regular.

$\text{pdim}_{Q/(f_a)}(K) = \infty$ (Auslander-Buchsbaum-Serre).

$$\Rightarrow V_R(K) = A_K^n.$$

- [Biggs - Gelfand - Prolitz, 25] Embedded deformation problem for monomial ideal.

- $n \leq 5$, [BGP, 25] classifies CSV of R , where $\mu(I) = n$ and I is a monomial ideal.

Setup: (Q, m, K) a RLR, $K = \bar{K}$

(r) regular sequence in Q .

• $R = Q/J$, $J = (f_1, \dots, f_n)$, each f_i a monomial in (r) .

Thm [BHP, 25]

If $n \leq 5$, then the CSV of R , $V_R(R)$ is

- either a Coordinate Subspace of A_K^5
- Union of coordinate subspaces of A_K^5
- A_K^5
- Union of two hyperplanes.

$$V_R(R) := \begin{cases} V(x_1 x_5) \\ \{ \underline{a} \in A_K^5 \mid a_i = 0 \ \forall i \in S \} \end{cases}$$

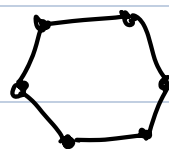
where $S \subseteq \{1, \dots, n\}$

$$x_1 + x_3 + x_5$$

Thm [FFKM SW, 26]:

If $n=6$, the following are realised as $V_R(R)$:

- a Coordinate Subspace of A_K^6 , with dimension $\neq 1$.
- $V(x_1, x_4 x_6)$
- $V(x_4 x_6)$
- $V(x_4 x_6, x_5 x_6)$
- $V(x_1, x_4 x_6)$.
- $V(x_1 x_3 x_5 + x_2 x_4 x_6)$



[hint 3, 26]

Question: ring defined by edge of an n -cycle
 $n \equiv 2 \pmod{4}$.

Thm: If $R = K[x_1, \dots, x_n]$
 $(x_2x_1, x_2x_3, \dots, x_nx_1)$

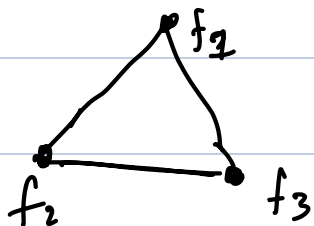
Then,
$$\mathbb{V}_R(R) = \begin{cases} \mathbb{V}(x_1x_3 \dots x_{n-1} + x_2x_4 \dots x_n) & \text{if } n \equiv 2 \pmod{4} \\ A_K^n & \text{otherwise.} \end{cases}$$

Strategy: $\mathbb{V}_R(R)$
• as $\mathbb{V}(I_{2^n-1}(M_J))$
 M_J - adjacency matrix.

- Let G_J - GCD graph of J consists of:
 - Vertices - $\{f_1, \dots, f_n\}$
 - edges - $\{f_i - f_j \mid \gcd(f_i, f_j) \neq 1\}$.

Ex: $J = (xy, xz, yzw) \subseteq K[x, y, z, w]$

$x, y, z, w \neq 1$



T_J - Taylor graph of J .

- $T_J = (V, E_d \cup E_n)$.

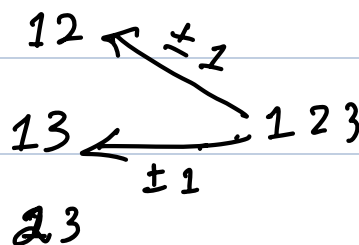
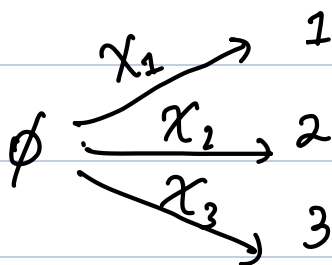
- $V = 2^{\{f_1, \dots, f_n\}}$

$$A \subseteq \{f_1, \dots, f_n\}$$

- $E_d = \{ A \xrightarrow{\pm 1} A \setminus \{f_i\} \mid \text{lcm}(A) = \text{lcm}(A \setminus \{f_i\}) \}$
(diff edges).

- $E_J = \{ A \xrightarrow{\pm x_i} A \cup \{f_i\} \mid f_i \notin N_{G_J}(A) \}$
(homotopy edges).

Ex: $J = (xy, xz, yzw)$



defn: M_J - adjacency matrix of J . $(2^n \times 2^n)$

↓

Targets

→

Sources

Weights

	ϕ	12	13	23	1	2	3	123
ϕ								0
12								±1
13								±1
14								0
1	x ₁							
2	x ₂							
3	x ₃							
123	0							

→

$A^3_k \ni \underline{a} = (a_1, a_2, a_3) \rightarrow$

$$\mathbb{A}_K^3 \ni \underline{a} = (a_1, a_2, a_3) \rightarrow$$



- Taylor resolution for monomial ideals
- [Eisenbud - Shamash] construction
- System of higher homotopies
- Two-periodic complex with M_J 's as differentials

$$\underline{a} \in V_R(R) \iff M_J(\underline{a}) \text{ gives an exact complex}$$

[BGP] - SL math notes

- Embedded deformation paper.

$$V_R(R) = \left\{ (a_1, \dots, a_n) \in \mathbb{A}_K^n \mid \text{rank } M_J \leq 2^{n-1} \text{ when } x_i = a_i \right\}$$

$$\text{So, } I(V_R(R)) = \sqrt{I_{2^{n-1}}(M_J)}$$

Ex: $I_{2^2}(M_J) = (0) \Rightarrow V_R(R) = \mathbb{A}_K^3.$

Ongoing [Stephen Stern] :

$n = 7, 8$ $\boxed{n=7}$ monomial ideals.

- talk with Stephen

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